Karl Weisenburger

Department of Mathematics, 150 N. University St, West Lafayette, IN 47907
kweisen@purdue.edu • math.purdue.edu/~kweisen
LinkedIn • GitHub
U.S. Citizen

EDUCATION

Ph.D. Candidate in Mathematics **Aug 2022 – Present**
Purdue University (West Lafayette, IN)

Advisors: Gregery Buzzard & Charles Bouman

Areas: Computational imaging, tomography, Bayesian inverse problems

Expected: Fall 2027

B.S. in Applied Mathematics, Minor in Latin **2018 – 2022**
Hillsdale College (Hillsdale, MI)

Honors: Summa Cum Laude, GPA 3.95/4.0

RESEARCH EXPERIENCE

Graduate Research Assistant **May 2023 – Present**
Purdue University (West Lafayette, IN)

Advisors: Gregery Buzzard, Charles Bouman (Purdue)

Collaborator: Matthew Kemnetz (AFIT)

- Developed **WindDensity-MBIR**, a model-based iterative reconstruction method that recovers high-order wavefront features to within 10–25% error in severely underdetermined regimes (sparse views, narrow field of view, limited angular extent) — even with piston, tip, and tilt removed from the measurements. Published in *Optical Engineering*.
- Built the Bayesian inverse-problem framework underlying these reconstructions for sparse, limited-angle wind-tunnel measurement data, in collaboration with the Air Force Research Laboratory.

Doctoral Research Fellow (ORISE) **May – Aug 2026**
Air Force Institute of Technology (Virtual)

Mentor: Matthew Kemnetz

- Developed a single-shot, limited-angle tomographic method for estimating wavefront tip and tilt from higher-order wavefront data.

AFRL Scholars Program Summer Intern **Summers 2024, 2025**
Air Force Research Laboratory (Virtual)

Mentor: Matthew Kemnetz

- Built simulation and reconstruction software and mathematical models for tomographic wavefront sensing; results were published in SPIE proceedings and presented at SPIE Optics + Photonics.

Data Analytics Intern **May – Aug 2022**
Build Central (Virtual)

- Trained machine-learning models to predict construction project completion dates and built analytics dashboards used by leadership to productize the results.

Laureates Honors Program Researcher **2021 – 2022**
Hillsdale College (Hillsdale, MI)

- Developed sparse, low-rank decomposition methods to denoise reconstructed neuronal connectivity matrices.

PUBLICATIONS

Refereed Journal

- **K. J. Weisenburger**, G. T. Buzzard, C. A. Bouman, M. R. Kemnetz, "WindDensity-MBIR: Model-based iterative reconstruction for wind tunnel 3D density estimation," *Optical Engineering* **65**(8), 081820 (2026). `https://doi.org/10.1117/1.OE.65.8.081820`

Conference Proceedings

- **K. J. Weisenburger**, G. T. Buzzard, C. A. Bouman, M. R. Kemnetz, "Tomographic wavefront sensing simulations for wind tunnel measurements," *Proc. SPIE 13619, Unconventional Imaging, Sensing, and Adaptive Optics 2025*, 136190R (2025). `https://doi.org/10.1117/12.3063657`

SOFTWARE

WindDensity-MBIR — open-source Python implementation of the model-based iterative reconstruction method for wind-tunnel 3D density estimation; reproduces all results and figures from the *Optical Engineering* paper.
github.com/Karl-Weisenburger/WindDensity-MBIR

PRESENTATIONS

- *Model-Based Iterative Reconstruction for Wind Tunnel 3D Wavefront Estimation*, DE Student Education Workshop, Directed Energy Professional Society (DEPS) Annual DE Science & Technology Symposium (Mar–Apr 2026).
- *Wind Tunnel Tomographic Wavefront Sensing Simulations*, IS&T Electronic Imaging Symposium, Burlingame, CA (Mar 2026).
- *Tomographic Wavefront Sensing Simulations for Wind Tunnel Measurements*, SPIE Optics + Photonics, San Diego, CA (Aug 2025).

- *Tomographic Wavefront Sensing Simulations for Wind Tunnel Measurements*, Directed Energy Professional Society (DEPS) S&T Symposium (May 2025).
- *Algorithmic Techniques for Interpreting the Reconstructed Connectivity Matrix*, Hillsdale College Math Department (Apr 2022).
- *Harmonic Measure Distribution Functions and Circle Domains*, Michigan MAA Spring Conference (Apr 2021).

TECHNICAL SKILLS

Languages & Tools: Python (JAX, NumPy, SciPy, PyTorch), MATLAB, C++, LaTeX, Git.

High-Performance Computing: Slurm-managed HPC clusters (Purdue); large-scale parallel workflows.

Methods: Bayesian inverse problems, computational tomography, model-based iterative reconstruction (MBIR), sparse / limited-angle reconstruction, numerical optimization.

AWARDS & HONORS

Member, Laureates Honors Program **2021 – 2022**
Hillsdale College (Hillsdale, MI)

- Awarded competitive research funding as one of 20 selected students; completed an independent project in computational neuroscience.

Ruth L. Schulz Memorial Mathematics Prize **Feb 2022**
Hillsdale College (Hillsdale, MI)

TEACHING

Graduate Teaching Assistant, *Purdue University* **2022 – 2023**

Teaching Assistant & Math Tutor, *Hillsdale College* **2021 – 2022**

LEADERSHIP & SERVICE

President **2020 – 2021**
Hillsdale College Catholic Society (Hillsdale, MI)

- Recruited and managed three student boards; organized speaker events, lecture series, and retreats.

ESTEEM Mentor **2025 – 2026**
St. Thomas Aquinas Community Parish (West Lafayette, IN)